



RESEARCH & INNOVATION UPDATES

Technologies for off-road autonomous driving



Researcher Han-Min Lee of the KIMM giving an explanation of the research outcome related to the newly developed technologies (Credit: KIMM)

Researchers at Korea Institute of Machinery and Materials (KIMM) have devised technologies tailored for navigating off-road terrains like mountains, watersides, or snowy areas. These advancements encompass sensor protection and cleaning, sensor signal correction, and recognition of drivable areas, all now adopted by relevant industries. The sensor protection module ensures sensors stay clean by promptly wiping off substances like mud or muddy water that may splatter during off-road journeys. Meanwhile, the signal correction tech removes minor particles like dust or snow, ensuring stable driving even in harsh weather. Moreover, their drivable area estimation tool detects obstacles and automatically charts alternative routes, crucial for avoiding collisions. These innovations greatly enhance off-road driving safety and efficiency, addressing previous gaps in sensor protection and real-time obstacle detection. With improved processing speed and performance metrics, these technologies are poised for practical application in autonomous off-road driving systems.

Breakthrough in making sodium batteries commercially

Researchers from Osaka Metropolitan University have introduced a method for making sodium-containing sulphides using sodium polysulphides. This method acts as both material and flux, leading to a solid sulphide electrolyte. This electrolyte has the highest sodium ion conductivity reported so far, around ten times higher than what is usually needed for practical use. Additionally, they have created a glass electrolyte with excellent resistance

to reduction. This breakthrough is vital for making all-solid-state sodium batteries commercially viable, which are seen as safer and potentially cheaper than lithium-ion batteries due to abundant sodium. The team believes their process could become standard for future sodium battery materials, offering sustainable energy storage solutions for a greener future.